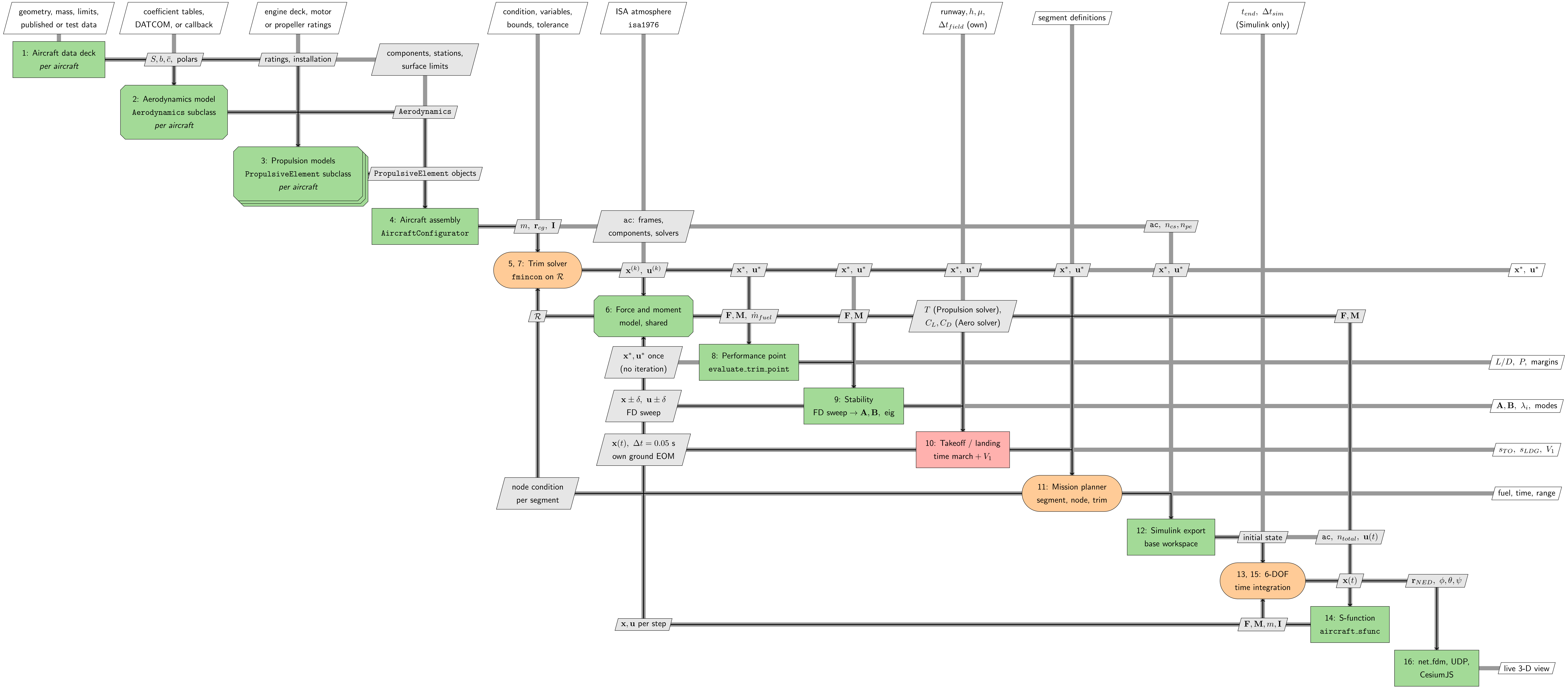
geometry, mass, limits, published or test data
coefficient tables, DATCOM, or callback
engine deck, motor or propeller ratings
condition, variables, bounds, tolerance
ISA atmosphere isa1976
runway, h, μ, Δt_{field} (own)
segment definitions
t_{end}, Δt_{sim} (Simulink only)
1: Aircraft data deck *per aircraft*
$S, b, \bar{c}$, polars
ratings, installation
components, stations, surface limits
2: Aerodynamics model Aerodynamics subclass *per aircraft*
Aerodynamics
3: Propulsion models PropulsiveElement subclass *per aircraft*
PropulsiveElement objects
4: Aircraft assembly AircraftConfigurator
m, $\mathbf{r}_{cg}$, $\mathbf{I}$
ac: frames, components, solvers
ac, n_{cs}, n_{pe}
5, 7: Trim solver fmincon on $\mathcal{R}$
$\mathbf{x}^{(k)}$, $\mathbf{u}^{(k)}$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathcal{R}$
6: Force and moment model, shared
$\mathbf{F}, \mathbf{M}$, $\dot{m}_{fuel}$
$\mathbf{F}, \mathbf{M}$
T (Propulsion solver), C_L, C_D (Aero solver)
$\mathbf{F}, \mathbf{M}$
$\mathbf{x}^*, \mathbf{u}^*$ once (no iteration)
8: Performance point evaluate_trim_point
L/D, P, margins
$\mathbf{x} \pm \delta$, $\mathbf{u} \pm \delta$ FD sweep
9: Stability FD sweep → $\mathbf{A}, \mathbf{B}$, eig
$\mathbf{A}, \mathbf{B}$, λ_i, modes
$\mathbf{x}(t)$, $\Delta t = 0.05$ s own ground EOM
10: Takeoff / landing time march + V_1
s_{TO}, s_{LDG}, V_1
node condition per segment
11: Mission planner segment, node, trim
fuel, time, range
12: Simulink export base workspace
initial state
ac, n_{total}, $\mathbf{u}(t)$
13, 15: 6-DOF time integration
$\mathbf{x}(t)$
$\mathbf{r}_{NED}$, ϕ, θ, ψ
$\mathbf{x}, \mathbf{u}$ per step
$\mathbf{F}, \mathbf{M}, m, \mathbf{I}$
14: S-function aircraft_sfunc
16: net_fdm, UDP, CesiumJS
live 3-D view